

Table B: Task B — Directory Assignment (Dir Acc@k)

Model	Method	Repr	Best Layer	Dir Acc@1	Dir Acc@3
LaTa	Baseline	hidden	12	0.7261	0.9139
LaTa	Baseline	ff1	1	0.5227	0.7512
LaTa	SIF only	hidden	1	0.8294	0.9531
LaTa	SIF only	ff1	1	0.8451	0.9531
LaTa	SIF+ABTT (opt D)	hidden	12	0.8592	0.9468
LaTa	SIF+ABTT (opt D)	ff1	6	0.8419	0.9390
LaTa	Whitening	hidden	4	0.0767	0.0861
LaTa	Whitening	ff1	9	0.1049	0.1142
PhilTa	Baseline	hidden	12	0.6244	0.8482
PhilTa	Baseline	ff1	2	0.3568	0.4632
PhilTa	SIF only	hidden	1	0.8044	0.9437
PhilTa	SIF only	ff1	8	0.7684	0.9327
PhilTa	SIF+ABTT (opt D)	hidden	9	0.8310	0.9437
PhilTa	SIF+ABTT (opt D)	ff1	8	0.8466	0.9562
PhilTa	Whitening	hidden	12	0.0689	0.0829
PhilTa	Whitening	ff1	12	0.1095	0.1205
LaBSE	Baseline	hidden	11	0.8357	0.9374
LaBSE	SIF only	hidden	12	0.8326	0.9405
LaBSE	SIF+ABTT (opt D)	hidden	9	0.8169	0.9311
LaBSE	Whitening	hidden	10	0.0657	0.0782
Qwen3-0.6B	Baseline	hidden	23	0.7872	0.9202
Qwen3-0.6B	Baseline	ffn_int	20	0.8169	0.9405
Qwen3-0.6B	SIF only	hidden	27	0.7606	0.9218
Qwen3-0.6B	SIF only	ffn_int	20	0.7746	0.9327
Qwen3-0.6B	SIF+ABTT (opt D)	hidden	26	0.7919	0.9264
Qwen3-0.6B	SIF+ABTT (opt D)	ffn_int	3	0.7840	0.9327
Qwen3-0.6B	Whitening	hidden	23	0.0876	0.1017
Qwen3-0.6B	Whitening	ffn_int	1	0.0814	0.0923
KaLM-mini	Baseline	hidden	22	0.8357	0.9452
KaLM-mini	Baseline	ffn_int	19	0.7919	0.9045
KaLM-mini	SIF only	hidden	24	0.8247	0.9327
KaLM-mini	SIF only	ffn_int	19	0.8106	0.9296
KaLM-mini	SIF+ABTT (opt D)	hidden	22	0.8122	0.9343
KaLM-mini	SIF+ABTT (opt D)	ffn_int	2	0.7700	0.9437
KaLM-mini	Whitening	hidden	5	0.0845	0.0955
KaLM-mini	Whitening	ffn_int	2	0.0876	0.0986